3.10.1.(2) $\left(\begin{array}{cccc|c}1&2&-3&-2&-6\\-2&-5&8&3&14\\1&2&-5&0&0\\1&2&-2&-3&-9\end{array}\right)\Rightarrow\left(\begin{array}{cccc|c}1&2&-3&-2&-6\\0&-1&2&-1&2\\0&0&-2&2&6\\0&0&1&-1&-3\end{array}\right)\Rightarrow\left(\begin{array}{cccc|c}1&0&-1&-4&\cancel{3}14\\0&-1&0&1&8\\0&0&-2&2&6\\0&0&0&0&0\end{array}\right)$

$\Rightarrow\left(\begin{array}{cccc|c|c}1&&&\cancel{-3}&[illegible]&1\\&-1&&1&8&8\\&&-1&1&3&3\\&&&&&\end{array}\right)$

故特解 $y=\cancel{\left(\begin{array}{c}5\ -1\\-8\\-3\\0\end{array}\right)}\left(\begin{array}{c}1\\-8\\-3\\0\end{array}\right)$ ~~而~~ 基础解系为 ~~$\eta=\left(\begin{array}{c}3\\0\\-1\\1\end{array}\right)$~~ $\eta=\left(\begin{array}{c}3\\1\\1\\1\end{array}\right)$

∴ 解为 $y+k\eta$

(4) $\left(\begin{array}{ccccc|c}1&2&1&-1&0&2\\2&4&3&-3&-1&5\\-1&-1&-3&3&2&-2\end{array}\right)\Rightarrow\left(\begin{array}{ccccc|c}1&2&1&-1&0&2\\0&0&1&-1&-1&1\\0&1&-2&2&2&0\end{array}\right)\Rightarrow\left(\begin{array}{ccccc|c}1&2&1&-1&&2\\&&1&-1&-1&1\\&1&&&&2\end{array}\right)$

$\Rightarrow\left(\begin{array}{ccccc|c}1&&1&-1&&-2\\&&1&-1&-1&1\\&1&&&&2\end{array}\right)\Rightarrow\left(\begin{array}{ccccc|c}1&&1&-1&&-2\\-1&[illegible]&&&-1&3\\&1&&&&2\end{array}\right)\Rightarrow\left(\begin{array}{ccccc|c}&&1&-1&-1&1\\1&&&&1&-3\\&1&&&&2\end{array}\right)$

∴ ~~基础为~~ 特解为 $y=\left(\begin{array}{c}-3\\2\\1\\0\\0\end{array}\right)$ 基础解系为 $\eta_1=\left(\begin{array}{c}0\\0\\1\\1\\0\end{array}\right)$ $\eta_2=\left(\begin{array}{c}-1\\0\\1\\0\\1\end{array}\right)$

∴ 解为 $y+k_1\eta_1+k_2\eta_2$

4、$\left(\begin{array}{ccccc|c}1&2&3&3&7&b\\3&2&1&1&a-5&b+2\\1&2&\cancel{8}0&\cancel{0}2&3a&b\\5&4&3&3&1-a&-b\end{array}\right)\Rightarrow\left(\begin{array}{ccccc|c}1&2&3&3&7&b\\&-4&-8&-8&a-2b&2-2b\\&&-3&\cancel{-3}-1&3a-7&0\\&-6&-12&-12&-a-3[illegible]&-6b\end{array}\right)\Rightarrow\left(\begin{array}{ccccc|c}1&2&3&3&7&b\\&4&8&8&2b-a&2b-2\\&&1&3&7-3a&0\\&&&&5-2.5a&[illegible]\end{array}\right)$

5、(1) 对 $\left(\begin{array}{cccc}1&2&0&5\\0&1&-2&4\\-1&0&1&-2\\3&1&1&4\end{array}\right)\Rightarrow\left(\begin{array}{cccc}1&2&0&5\\0&1&-2&4\\-1&0&1&-2\\3&0&3&0\end{array}\right)\Rightarrow\left(\begin{array}{cccc}1&2&0&5\\-2&1&0&0\\-1&0&1&-2\\3&0&3&0\end{array}\right)\Rightarrow$

$\left(\begin{array}{cccc}5&&&5\\-2&1&&\\-1&&1&-2\\1\cancel{8}&&1&\end{array}\right)\Rightarrow\left(\begin{array}{cccc}\cancel{5}1&&&\cancel{5}1\\-2&1&&\\-2&&0&-2\\1&&1&\end{array}\right)\Rightarrow\left(\begin{array}{cccc}1&0&0&1\\-2&1&0&0\\0&0&0&0\\1&0&1&0\end{array}\right)$

故 $k\alpha_1+k_2\alpha_2+k_3\alpha_3=\beta$ 有解，即 β 可用 $\alpha_1,\alpha_2,\alpha_3$ 表示

7 ~~6~~，若 $Ax=0$ 只有零解，则 $r(A)=\cancel{a}n$ ∴ $|A|\neq0$ ∴ $|A^k|\neq0$ ∴ $A^kx=0$ 只有零解

若 $A^kx=0$ 只有零解，则 $\cancel{|A^k}r(A^k)=\cancel{o}n$ ∴ $|A^k|\neq0$ ∴ $|A|\neq0$ ∴ $Ax=0$ 只有零解

8、对 $\left(\begin{array}{c}A\\\beta\end{array}\right)$，$\left(\begin{array}{c}A\\\beta\end{array}\right)x$ ~~x~~$=0$ 的解~~为~~与 $Ax=0$ 完全相同，故 $r\left(\begin{array}{c}A\\\beta\end{array}\right)=r(A)$

∴ ~~$\alpha_1,\alpha_2\dots\alpha_m$ 与 β 线性~~木 β 可由 $\alpha_1,\alpha_2\dots\alpha_m$ 线性表出

9、$Ax=\beta$ 有解 $\Leftrightarrow r(A)=r(A\vdots\beta)\Leftrightarrow r(A')=r\left(\begin{array}{c}A'\\\beta'\end{array}\right)\Leftrightarrow A'y=0$ 的适用于 $\alpha'\beta=0$

10、$\Rightarrow$ 若(1)有解，设 $A=(a_{ij})$ $B=\left(A\vdots\begin{array}{c}b_1\\\vdots\\b_n\end{array}\right)$ 则 $A=\left(\begin{array}{c}\alpha_1\\\alpha_2\\\vdots\\\alpha_n\end{array}\right)$ $B=(A\vdots\beta)$

则 $r(A)=r(B)$ ∴ $r(A')=r(B')$ 且 β 可由 $\alpha_1,\alpha_2\dots\alpha_n$ 线性表示

~~又~~：易知不存在 $k_1 0+\dots+k_n 0=1$，故 ~~$r\left(B'\vdots\begin{array}{c}0\\\vdots\\1\end{array}\right)=r+1$~~ $r\left(\begin{array}{c|c}A'&\begin{array}{c}0\\\vdots\\0\end{array}\\\beta'&1\end{array}\right)=r\left(A'\vdots\begin{array}{c}0\\\vdots\\0\end{array}\right)+1=r(A)+1$

∴ $r\left(\begin{array}{c|c}A'&\begin{array}{c}0\\\vdots\end{array}\\\beta'&1\end{array}\right)=r(A)+1\neq r(B')$ ∴ (2)无解

$\Leftarrow$ 若(2)无解 则 $r\left(B'\vdots\begin{array}{c}0\\\vdots\\0\\1\end{array}\right)\neq r(B')$ ∴ $r\left(B'\vdots\begin{array}{c}0\\\vdots\\0\\1\end{array}\right)=r(B')+1$

Recording start from hero

又∵ $r\begin{pmatrix} B' & \vdots & \begin{matrix}0\\ \vdots\\ 1\end{matrix}\end{pmatrix} = r\begin{pmatrix} A' & 0 \\ \beta' & 1 \end{pmatrix} = r(A' \vdots 0)+1 = 1+r(A') = r(A)+1$

∴ $r(A) = r(B') = r(B) = r(A \vdots \beta)$

∴ (1) 有解

11、由基础解系定义可知 $\alpha_1 \ldots \alpha_{n-r}$ 与 $\beta_1 \ldots \beta_{n-r}$ 是~~齐次~~同一个子空间的两组不同的基，且空间维数为 $n-r$，

设 P 与二者的过渡矩阵，则 $(\beta_1 \ldots \beta_{n-r}) = (\alpha_1 \ldots \alpha_{n-r})P$

12、~~对 $A = \begin{pmatrix} a_{11} & a_{12} & & \\ a_{21} & a_{22} & & \\ & & \ddots & \\ & & & a_{nn} \end{pmatrix}$ 通过基本列变化，变将第一行除 a_{11} 外变变为0~~

~~∵ $|a_{ii}| > \sum_{j=1, j\neq i}^{n} |a_{ij}|$ ∴ 变换后得到的 a_{ii}' 均不为0~~

对 $Ax=0$，若存在非零解 $(x_1, x_2 \ldots x_n)'$，

取 $x_m = \max\{|x_1|, \ldots |x_n|\}$，设 $x_m = |x_i|$

则对第 i 个方程 $x_i a_{ii} + (x_1 a_{i1} + \cdots + x_n a_{in}) = 0$ ①

∵ $|x_1 a_{i1}| + \cdots + |x_n a_{in}| - |x_i a_{ii}| < |x_i| \sum_{j=1}^{n} |a_{ij}| - |x_i a_{ii}| < 2|x_i a_{ii}|$

∴ ①不成立 ∴矛盾 ∴ $Ax=0$ 不存在非零解

∴ $r(A) = n$ ∴ A非奇

设 $f(t) = \begin{vmatrix} a_{11} & ta_{12} & & ta_{1n} \\ ta_{21} & a_{22} & & ta_{2n} \\ & & \ddots & \\ ta_{n1} & ta_{n2} & & a_{nn} \end{vmatrix}$ $(0 \leq t \leq 1)$ ∴ $f(0) = \prod_{i=1}^{n} a_{ii} > 0$ $f(1) = |A|$

且 $f(t)$ 为连续函数

由前知 $f(t) \neq 0$， ∴若 $|A| < 0$，则存有 $f(t_0) = 0$ 矛盾 ∴ $f(1) = |A| > 0$

13、∵ ~~存在 $A_{ij} \neq 0$~~ 若 A的阶 $r(A) \leq n-2$，则 A的任意 $n-1$ 阶子式为0

又∵存在 $A_{ij} \neq 0$ $|A| = 0$ ∴ $r(A) = n-1$ ∴ 故 A $Ax=0$ 的基础解系只有一个基

又∵对 $Ak\begin{pmatrix} A_{i1} \\ A_{in} \end{pmatrix}$，若第i行 $a_{i1}A_{i1} + a \ldots + a_{in}A_{in} = |A| = 0$

非第i行，由行列式性质得为0

∴ $A\left(k\begin{pmatrix} A_{i1} \\ A_{in} \end{pmatrix}\right) = kA\begin{pmatrix} A_{i1} \\ A_{in} \end{pmatrix} = 0$

综上，$Ax=0$ 的所有解为 $k\begin{pmatrix} A_{i1} \\ A_{in} \end{pmatrix} = 0$

14、①若 $r(A) \leq n-2$，由13，任意 $A_{ij} = 0$ ∴ $r(A^*) = 0$

②若 $r(A) = n-1$，由13，知存在 i，使 $Ax=0$ 的基础解系为 $k\begin{pmatrix} A_{i1} \\ A_{in} \end{pmatrix}$

且任意 $\begin{pmatrix} A_{j1} \\ A_{jn} \end{pmatrix}$ 都是 $Ax=0$ 的一组解

∴ $\begin{pmatrix} A_{j1} \\ A_{jn} \end{pmatrix} = k\begin{pmatrix} A_{i1} \\ A_{in} \end{pmatrix}$ ∴任意一个二阶子式 $\begin{pmatrix} a_{kj} & a_{kj+1} \\ a_{k+1j} & a_{k+1j+1} \end{pmatrix}$ 必有 $\begin{pmatrix} a_{kj} \\ a_{k+1j} \end{pmatrix} = k'\begin{pmatrix} a_{kj+1} \\ a_{k+1j+1} \end{pmatrix}$

∴ $\begin{vmatrix} a_{kj} & a_{kj+1} \\ a_{k+1j+1} & a_{k+1j+1} \end{vmatrix} = 0$ ∴ $r(A^*) \leq 1$

∴ $r(A^*) \leq 1$

Recording start from here

5、$\begin{pmatrix} 1 & 2 & 3 & 3 & 7 & b \\ 3 & 2 & 1 & 1 & a-5 & b+2 \\ 1 & 2 & 0 & 2 & 3a & b \\ 5 & 4 & 3 & 3 & 1-a & -b \end{pmatrix} \Rightarrow \begin{pmatrix} 0 & 0 & 0 & 0 & 5a-10 & 6b+6 \\ -3 & 0 & 0 & 2 & 16-a & -3b-6 \\ -2 & -1 & 0 & 0 & 8-2a & -2b-3 \\ 1 & 0 & 2 & 0 & 6-5a & -3b-2 \end{pmatrix}$

∴特解 $\begin{pmatrix} \frac{3ab+8a+22b+12}{5(a-2)} \\ \frac{-8ab-13a-16b-8}{5(a-2)} \\ \frac{6ab+6a-14b-14}{5(a-2)} \\ \frac{6b+6^{0}}{a-2} \end{pmatrix}$　　　齐基础解系 $\begin{pmatrix} 2 \\ -4 \\ -1 \\ 3 \\ 0 \end{pmatrix}$

若 $a=2$，$b\neq -1$，无解　　若 $a=2$，$b=-1$　$x_5=0$　$x_2=\frac{1}{2}-\frac{x_1}{2}$　$x_3=1-2x_1$　$x_4=-\frac{3}{2}+\frac{2}{3}x_1$

4.1.1 (1) 对 φ　$\varphi(\alpha+\beta)=k(\alpha+\beta)=k\alpha+k\beta=\varphi(\alpha)+\varphi(\beta)$.

$\varphi(c\alpha)=kc\alpha=ck\alpha=c\varphi(\alpha)$

∴ φ 是

(3) 对 φ　$\varphi(f(x)+g(x))=\int_0^x (f(t)+g(t))\,dt=\int_0^x f(t)\,dt+\int_0^x g(t)\,dt=\varphi(f(x))+\varphi(g(x))$

$\varphi(c\,f(x))=\int_0^x (c\,f(t))\,dt=c\int_0^x f(t)\,dt=c\,\varphi(f(x))$

∴ φ 是

(5) 不是　$\varphi(x_1+x_2, y_1+y_2)=(x_1+x_2+a, y_1+y_2+b)\neq(\overset{x_1+a,}{\cancel{x_1+x_2+2a}}\ y_1+b)+(x_2+a, y_2+b)=\varphi(\overset{x_1,y_1}{x_1})+\varphi(x_2,$

2、若 $\beta=0$，$\varphi(\alpha)=A\alpha$

∴ $\varphi(\alpha+\beta)=A(\alpha+\beta)=A\alpha+A\beta=\varphi(\alpha)+\varphi(\beta)$

$\varphi(c\alpha)=A(c\alpha)=cA\alpha=c\varphi(\alpha)$

∴ φ 是线性变化

若 φ 是线性变化

∴ $\varphi(2\alpha)=2\varphi(\alpha)$　∴ $2A\alpha+2\beta=2A\alpha+\beta$　∴ $\beta=0$

3. 对 $\forall\alpha\in V$，α 存在唯一的 α_1, $\alpha_1\in V_1$，$\alpha_2\in V_2$，使 $\alpha=\alpha_1+\alpha_2$

∴ $\varphi(\alpha)=\varphi_1(\alpha_1)+\varphi_2(\alpha_2)$ 满足条件

若存在 ψ 也满足条件

∴ $\psi(\alpha)=\psi(\alpha_1)+\psi(\alpha_2)=\varphi_1(\alpha_1)+\varphi_2(\alpha_2)=\varphi(\alpha)$

∴ $\psi=\varphi$，∴ φ 唯一

4. ① 先证线性变化　$\varphi(ca_0, ca_1, \dots ca_n \dots)=ca_0+ca_1x+ca_2x^2+\cdots+ca_nx^n=c\varphi(a_0, a_1, \dots a_n \dots$

$\varphi(a_0+b_0, a_1+b_1, \dots a_n+b_n \dots)=a_0+b_0+(a_1+b_1)x+\cdots+(a_n+b_n)x^n=\varphi(a_0, a_1 \dots a_n)+\varphi(b_0, b_1 \dots b_n)$

∴ φ 是线性变化

② 单设若 $a_0+a_1x+\cdots+a_nx^n=b_0+b_1x+\cdots+b_nx^n$ 则 $a_0=b_0$　$a_1=b_1 \dots$　$a_n=b_n \dots$

∴ $(a_0, a_1 \dots a_n \dots)=(b_0, b_1 \dots b_n \dots)$

Recording start from here

③满射，对任意 $a_0 + a_1x + a_2x^2 + \cdots + a_nx^n$

存在 $(a_0, a_1, \ldots, a_n, 0, 0, \ldots)$ 使 $\varphi(a_0, a_1, \ldots, a_n, 0, \ldots) = a_0 + a_1x + \cdots + a_nx^n$

综上，φ 是线性同构

4.2.2 设 $\varphi(\alpha) = A\alpha$ 则 $(k\varphi)(\alpha) = kA\alpha = kI_V(A\alpha) = (kI_V)\circ\varphi(\alpha)$

$= A\alpha(kI_V) = \varphi(kI_V(\alpha))$

$\therefore k\varphi = (kI_V)\circ\varphi = \varphi\circ(kI_V)$

4.3、$DS(f(x)) = D(\int_0^x f(t)\,dt) = \frac{d}{dx}(\int_0^x f(t)\,dt)$

由积分，导数定义知 S 是 $V \to V$ 是单射，非满射，D $V \to V$ 是满射，非单射

设 $S(f(x)) = F(x)$，~~则F(x)~~ $\therefore F$ $\frac{d}{dx}F(x) = f(x)$

$\therefore DS(f(x)) = f(x)$ $\therefore DS = I_V$

而对 $f(x) = 1$ $\frac{d}{dx}f(x) = 0$ $\therefore SD(f(x)) = \int_0^x 0\,dt = 0 \neq f(x)$

$\therefore SD \neq I_V$

Recording start from here